

R AKASH KUMAR

Bangalore, IND | +91 6363699022 | [linkedin](#) | [Leetcode](#) | [Github](#)

EDUCATION

R V College of Engineering, Bangalore 3.73/4 | 9.34/10.0
Bachelor of Engineering, Computer Science and Engineering

Bangalore, India
Sep 2023 - Sep 2027

RELEVANT EXPERIENCE

Project Intern — Dept. of CSE, RVCE | Bangalore

(Mar 2025 – Jun 2025)

Project: *BCI-Based Smart Home Automation System*

- Built a Flask-based web interface to display real-time EEG metrics processed via Neiry API
- Trained an ML model to classify mental states (fatigue, focus, relaxation) from EEG features
- Implemented user-consent-driven control flow to trigger Raspberry Pi GPIO for IoT devices via network API

PROJECTS

REAL-TIME CODE MENTOR WITH ERROR PREDICTION:

March 2026

- Built an AI-powered programming assistant that analyzes student code to deliver structured, multi-level hints — improving debugging skills and logical reasoning through guided learning.
- Integrated Monaco Editor for browser-based coding with sandboxed execution via Piston API and Groq LLM for AI-driven error analysis, generating contextual hints and step-by-step explanations.

AI-POWERED GRIEVANCE INTELLIGENCE & PREDICTIVE INFRASTRUCTURE RISK PLATFORM:

June 2026

- Developed an AI-powered smart city infrastructure monitoring platform analysing public grievance data to predict infrastructure failure hotspots and support proactive maintenance planning.
- Implemented geospatial heatmap visualization, NLP-based complaint intelligence, escalation risk scoring, and fraud anomaly detection using Random Forest & Isolation Forest — enabling predictive risk analysis and real-time dashboard analytics.

VOICE & GESTURE-BASED AUTOMATED SCOREKEEPING SYSTEM FOR BALL BADMINTON

Oct 2025

(Presented at International Sports Conference Delhi)

- Developed an AI-assisted automated scorekeeping system for ball badminton, addressing real-world scoring errors and service-hand confusion through intelligent automation.
- Built a web-based platform with voice and gesture-based score updates using speech recognition and MediaPipe hand tracking — enabling real-time scoreboard synchronization and automated match logging.

SKILLS

Technical: *Programming - C++, C, Python | Web Development - HTML, CSS, ReactJS, NodeJS, Flask | Database - SQL, MongoDB | Python Libraries- Numpy, Pandas, Matplotlib, Scikit-learn, Tensorflow*

POSITION OF RESPONSIBILITY

Kannada Sangha RVCE, Bangalore, KA. Executive Member, Fest Coordinator

Dec 2023 - Present

- Led event organization and sponsorship acquisition as Fest Coordinator; contributed to club operations as an Executive Member.

SPORTS, Ball Badminton-Team Captain

- Represented RVCE at Zonal and State-level tournaments, achieving Top 4 and Runner-Up finishes

CERTIFICATIONS

- NPTEL course on Data Science for Engineers, Decision Modelling - received Elite+ Silver Certificate
- **IBM Cognitive Class** certifications in Data Analysis with Python and Data Visualization with Python, and **Oracle Cloud Infrastructure** 2025 certifications in Foundations Associate and Generative AI Professional – certified by Oracle University.